RESQNET AI

Explainable multi-agent flood-rescue coordination and disaster-response simulation platform

Project in one sentence
RESQNET AI is a digital twin that simulates how autonomous emergency teams coordinate rescue resources, routes, and shelter capacity during floods and infrastructure failures.

Project overview

RESQNET AI creates a synthetic digital model of a flooded city. Rescue boats, ground teams, incident coordinators, and shelters act as cooperating agents. Together they decide which emergencies to serve, which safe routes to use, and where rescued people can be accommodated.

The platform is designed as a research and decision-support simulator. It helps study whether decentralized coordination can remain effective, safe, fair, reproducible, and explainable when disaster conditions change quickly.

The problem being addressed

Flood response teams must make urgent decisions while facing:

- Multiple SOS reports arriving at the same time
- Limited rescue boats, ground vehicles, and shelter capacity
- Flooded roads, bridge closures, and changing route conditions
- Delayed, duplicated, contradictory, or false reports
- Communication blackouts and incomplete local information
- Sudden equipment, infrastructure, and shelter failures

How coordination works

Every emergency becomes an incident. Available agents evaluate incidents using travel time, route risk, victim urgency, shelter availability, information reliability, and fairness between neighbourhoods. In the decentralized strategy, agents bid for suitable incidents; incident agents compare proposals; shelter capacity is reserved; and an independent safety layer validates every commitment.

If a route closes, a resource fails, or a shelter loses capacity, affected assignments are reconsidered and agents negotiate an alternative plan.

Core capabilities

Live command centre	City map, flood conditions, incidents, rescue units, shelters, assignments, activity feeds, and performance indicators.
Multi-agent coordination	Contract-net bidding, deferred acceptance, route planning, and shelter reservations.
Strategy comparison	Decentralized, centralized, and greedy coordination can be tested under identical conditions.
Scenario simulation	Normal flood, flash flood, bridge closure, resource shortage, shelter overflow, conflicting information, and triple failure.
Safety shield	Blocks unsafe routes, double-booking, excessive loads, stale decisions, invalid reservations, and disabled resources.
Explainable decisions	Shows why an agent was selected, which alternatives were rejected, and which safety checks were applied.
Replay and audit	Records events and decisions for inspection and deterministic reconstruction of a run.
Decision lab and benchmarks	Supports counterfactual experiments, adversarial failures, repeated trials, and comparative reports.

Technology architecture

Frontend	React, TypeScript, Vite, Zustand, Recharts, React Flow, Tailwind CSS
Backend	Python and FastAPI
Live updates	WebSockets
Persistence	SQLite and event-sourced simulation state

Project objective

The central goal is to demonstrate and evaluate transparent multi-agent decision-making for emergency coordination. The system provides a controlled environment for analysing response time, rescue outcomes, unmet demand, resource use, fairness, robustness, and safety across difficult disaster scenarios.

"RESQNET AI is an explainable multi-agent digital twin for testing safe, adaptive, and fair flood-rescue coordination."

Current scope: This implementation uses synthetic simulation data. It is not yet connected to live emergency services, field sensors, or real-world dispatch systems.